

CHAPTER 4

The Human Body



Look at yourself in a mirror. Turn around at different angles while looking at your reflection. What do you see? Do you wonder how every part of your body works?

The human body consists of many parts. These parts work together to keep the body functioning. Some of the parts are cells, tissues, organs, and organ systems. Each part has a specific structure and function. For example, the heart works tirelessly to pump blood which goes around inside the body so that every cell is supplied with oxygen and waste products are removed before they can harm the body. The brain connects and controls all the parts of the body, and the interaction among these parts allows the body to function well. The human body is truly amazing.

In this chapter, you will learn about the parts of the body and their specific roles.

- What are the different organs that make up each of your body systems? What are the functions of each?
- What are the functions of each of your body systems and how are these functions performed?
- How do the organs of each body system work together to ensure that the body system is able to function well?
- How do the different body systems work together to allow one another to perform their roles effectively?
- How do you care for your body systems?
- Why is it necessary to learn about the parts and functions of each body system and their interactions?

The digestive system consists of the mouth, esophagus, stomach, small intestine, large intestine, and anus. There are three related organs involved in digesting food. These are the liver, pancreas, and gallbladder.

The parts of the digestive system work together to perform the functions of the entire system. Although each part has its own specific functions, their interaction allows the proper digestion of the food that enters the body.

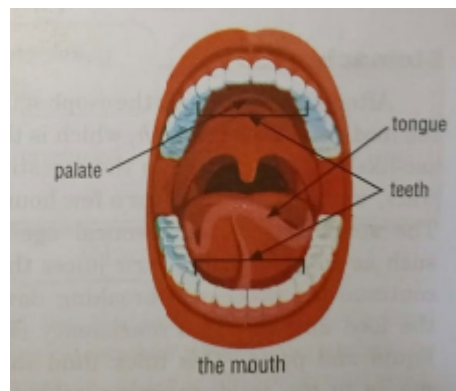
Food can be digested by a combination of two methods: mechanical digestion and chemical digestion. Mechanical digestion involves the physical breaking down of food into smaller pieces by chewing or grinding food with the help of the teeth. However, chewing is not enough to break down food into parts that can be absorbed by cells. As you chew, saliva in your mouth starts off chemical digestion, which involves the secretion of chemical agents that break the bonds that hold the particles of food together. After proteins, carbohydrates, and other nutrients in food react with digestive enzymes, these are broken into small digestible parts and absorbed by the cells in the intestines. Digestion ends with the movement of waste materials through the rectum and anus.

Mechanical and chemical digestion take place in the digestive tract or alimentary canal, the continuous muscular tube which carries food through different parts of the digestive system. There are several organs that are part of the alimentary canal. These organs work together to break down food and manage waste materials.

Parts of the Digestive System and Their Functions

Mouth

The mouth is the oral cavity and the first part of the alimentary canal. It contains the teeth, tongue, and palate. It is the entry point for food. The digestive system prepares for the entry of food in the mouth by releasing saliva at the sight and smell of food. Once food enters the mouth, digestion begins. Through mastication, which is the process of grinding and chewing with the help of the teeth, food is separated into smaller pieces and broken down further by the enzymes



in the saliva. Saliva

keeps the mouth moist and lubricates the food. This helps the tongue turn the food into a soft rounded mass called a bolus, which is ready for swallowing and passage into the pharynx. When swallowing happens, the palate, which is the roof of the mouth, serves as a barrier between the mouth and the nose so that food does not leak back to the nose.

Extend Your Understanding Word Unlock

Care for the Musculoskeletal System

When something goes wrong with the muscles, bones, or joints, pain and discomfort are felt by the body. To prevent this, you must take care of these parts of your body. You need a healthy musculoskeletal system to move, lift objects, and do exercises. How can you take good care of this system?

Exercising

Bones need resistance training to increase bone density, Muscles need to be worked out maintain a healthy musculoskeletal system. Yoga exercises can also increase strength and flexibility and maintain proper posture.

running and yoga exercises

Getting the Right Food and Nutrition

Eating food rich in calcium helps develop strong and healthy bones. You can get calcium from leafy vegetables, sardines, milk, and other dairy products. Like calcium, vitamin D also plays a vital role in the musculoskeletal system. Vitamin D increases the absorption of calcium into the bones.

foods rich in calcium

Getting Enough Sleep

At your age, you should get at least eight hours of sleep every night. This is important because your body repairs broken tissues while you sleep. Muscles relax during sleep. While you sleep, your brain also clears and restores itself, allowing to help relieve tension and reduce symptoms of sickness. Muscle growth also occurs you to feel alert and refreshed when you wake up.

Extend Your Understanding (Know Zone)

Did you know that...

> nicotine, a component found in cigarettes, and alcohol make bones weak and brittle by removing calcium from bone cells?

poor sleep puts you at risk of serious medical conditions and shortens your life expectancy? stress and overuse of computers, phones, and other gadgets have profound effects on your heart?

The musculoskeletal system is composed of bones, joints, and muscles.

A bone is a living organ that is made up mostly of collagen and calcium phosphate.

Bones are classified according to types of tissue as cortical bone or cancellous bone.

There are five types of bones: flat, long, short, irregular, and sesamoid. The joint is the connection between the bones.

Joints are classified according to the degree of their movement. The types are fibrous (immovable), cartilaginous (partially movable), and synovial (freely movable).

Muscles are specialized tissues that have the ability to contract and effect movement.

The muscular system has three types of muscles: skeletal, smooth, and cardiac.

Some ways of caring for the musculoskeletal system are exercising.

getting the right food and nutrition, and getting enough sleep.

The muscles and the skeleton function to protect and provide support to the other organs in the body. The skeleton keeps the body stable. Without the skeleton, the soft organs of the other body systems will be prone to damage or dysfunction. The muscles, in turn, allow movement by keeping the bones in place. When muscles contract, the bones attached to the joints also move. Diseases of the body affect every part of it. For example, influenza does not only affect the respiratory system but also the muscular system. This is why your muscles ache and you feel weak when you are sick with the flu.

Do you know how the bones and muscles are arranged? Do the next activity to see how these structures are connected.

Ramoc.

Explore Around

INVESTIGATING BONES AND MUSCLES

What You Need

chicken wings

water

vinegar

cup or bowl

What to Do

paper towels

illustration board

glue gun and glue stick

1. Wash the chicken wings and carefully remove the skin. Locate the biceps and triceps and show them to your teacher.
2. Remove all the flesh and clean the bones. Then soak them in vinegar in a cup or bowl for 5 minutes.
3. After 5 minutes, dry the bones with paper towels and assemble them on the illustration board.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. In what way are the chicken bones similar to human bones?
2. What do you think are the functions of the muscles and bones in the wings?

How You Learned

1. Did you have any difficulty in doing the activity? What are these difficulties?

2. Were you able to locate the muscles and bones easily? How did you do it?

Express Your Learning

Answer the following questions:

1. What are the types of muscles? How do they differ from one another?
2. What are the functions of bones?
3. How can you take good care of your musculoskeletal system?
4. How do the muscles and bones work together to form the musculoskeletal system?
5. How did you feel about the lesson? Explain your answer.

Lesson 2

Integumentary System

Do List

1. Name the parts of the integumentary system and their functions.
2. Identify the functions of the integumentary system.
3. Explain how the organs of the integumentary system work together with other organ systems.
4. Enumerate ways of taking care of the integumentary system.

Energize Your Mind



Look at your skin and feel its texture. Look at your hair. Have you ever wondered Why does your hair grow long? What about your nails? Look at the pictures above. A young man is fixing his hair, and the young lady is feeling her arm. These teenagers can feel sensations while combing their hair and touching their skin because the skin sends signals to the brain. Just like the two young people in the picture, you can also feel certain characteristics of objects, such as their texture or shape, even if your eyes are closed. Why is this so? What are the parts of the skin that make you feel things in the environment? Is there a relationship between your brain and what your skin feels?

Do the following activity to have an idea of how your skin functions

Explore Around

What You Need

TOUCH LAB

paper clips

Pen

What to Do

1. Bend a paper clip into a V or U shape.

2. Hold the paper clip on its bent end while pressing the tips onto your finger.

Press using one tip only, and then do it again using two tips. Observe the sensation that you feel on your fingers. Rate your sensitivity from 1 to 3, with 3 being the most sensitive. Record your ratings in the corresponding columns in the table below.

3. Repeat step 2 but this time, press the paper clip onto your palm and then on your forearm. Record your ratings in the table.

4. Do the steps again with your eyes closed.

5. Get a partner and repeat the procedures on him/her. Record his/her ratings in the same table.

Sensitivity of skin

Name

1 tip

Fingers

Sensitivity to the Paper Clip Tips

Opened Eyes

Closed Eyes

Opened Eyes

2 tips

Palm

Closed Eyes

1 tip

Opened Eyes

Closed Eyes

2 tips

Forearm

Opened Eyes

Closed Eyes

Opened Eyes

1 tip

Closed Eyes

2 tips

Opened Eyes

Closed Eyer

What You Have Learned

Answer the following questions. Write your answers in the space provided.

What does this imply?

1. Did you still feel the tip(s) of the paper clip when your eyes were closed?
2. Which part (finger, palm, or forearm) was the most sensitive to touch? Why do you think this part is the most sensitive?
3. How do you compare your responses to touch with those of your partner?

How You Learned

1. Did you encounter any problems in doing the activity? What did you do to resolve it?
2. Was the experience of doing the activity different with a partner?

Equip Yourself

In the activity, you have observed how the skin feels and responds to sensations caused by different objects.

The skin acted as a barrier that prevented the paper clips from causing stronger sensations or even harm on the parts beneath it. The skin is one of the parts of an organ system called the integumentary system. The integumentary system is a set of organs that provides external bodily covering. Its main function is to protect everything in the body from any harm.

Parts and Functions of the Integumentary System

Each part of the integumentary system has a specific function.

Skin

The skin is an organ that is made up of a layer of tissues that protect underlying organs. While it absorbs nutrients from the sun and other natural elements, it also protects the body from pathogens or disease-causing microorganisms in the environment. It is thus considered the first line of defense that protects the body from the environment. It also functions to prevent the loss of body fluids, protect against disease, eliminate waste materials, regulate body temperature, and give the sensations of touch, heat, and cold.

The exterior surface of the skin that covers the entire body has the largest area of all the organs. Thus, the skin is considered the largest organ of the body.

The skin is divided into different layers.

hair

capillaries

sweat pore

↓

epidermis

sensory nerve

motor nerve

sweat gland

dermis

sebaceous gland

hair blue

thermoreceptor

hypodermis

hair follicle

layers and structure of the skin

The main layers of the skin are the epidermis, dermis, and hypodermis. The epidermis is the outermost layer of the skin that serves as a protective barrier. It is made up of different cells, which include keratinocytes and melanocytes. Keratinocytes produce keratin, the protein that makes up the hair, nails, and outer layer of skin and protects the skin and tissue. Melanocytes produce melanin, a pigment that determines skin color. The more melanin in the skin, the darker the color. In this layer, skin cells are produced, sloughed away, and replaced. This process takes place unnoticed every second of every single day.

1. The dermis is the layer of skin that lies beneath the epidermis. It is the thickest layer of the skin and is made up of fibrous and elastic tissue. It contains structures that include the sweat glands, blood vessels, oil glands, and nerves. The main function of the dermis is to support the epidermis and enable the skin to thrive.

Hair

hair shaft

hair bulb

hair root

hair cell

Hair, the threadlike outgrowths from follicles that are found in the dermis, is made of tough proteins called keratin, which protects the hair from dryness and damage, and melanin, which gives hair its color. Almost every part of the body is covered with hair. A hair follicle anchors each strand of hair to the skin. In the hair bulb at the base of the hair follicle, cells divide and grow to build the hair shaft. Blood vessels provide nourishment to the cells in the hair bulb. Humans have three types of hair: lanugo, vellus, and terminal hair. Lanugo hair is fine hair that covers nearly the entire body of the fetus. By the time of the baby's birth, this hair is replaced by vellus. Vellus hair is the short, thin body hair that develops during childhood. Terminal hair is fully developed hair that is influenced by the hormonal changes in puberty and is found on the scalp, face, armpits, and pubic area.

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Hair grows in cycles of three phases. During the growth phase, most hair grows at any given time for a period of several years. In the transitional phase, hair growth slows down and the hair follicle begins to shrink. The resting period sets in, when hair growth stops and the old hair detaches from the hair follicle and falls out. Soon, a new strand of hair grows, and the growth phase begins anew.

The rate by which hair grows varies among individuals. As people age, the pigment cells die, and hair turns gray.

Extend Your Understanding

What causes goosebumps?

Know Zone

At the base of every strand of hair are small muscles known as arrector pills. Conditions such as flash-cold temperatures, the sound of scratches on the chalkboard, pleasant music, the sight of breathtaking places, intense emotions, coming face-to-face with iconic characters, and watching horror movies cause these muscles to contract and the hair strands to stand erect

Nails

all prate

A nail is a keratinous covering for the tips of the fingers and toes. Nails are often used for scratching and for doing many other tasks. Your nails help you grasp and hold small things. Nails also increase the sensitivity of the fingertips and protect them from damage. While nails seem to be basic parts of the hands and feet, they consist of a complex structure.

1001

cities

Nails have various parts: the nail

20

plate, nail head, cuticle, nail fold, lunula and the root. The nail plate is the visible part of the nail. The nail bed is the skin beneath the nail plate. The cuffole is the tissue at the base of the nails. The noil fold is the tissue that attaches the nail to the ship through the cuticle. It is the white area at the base of a nail. The root is the hidden part of the nail under the cuticle

The nails originate from epidermal cells below the root. These cells produce the keratin that makes the nail body strong and tough. These keratin-filled cells move away from the root and accumulate on the surface of the skin. As this happens, the cells become tightly pressed together, farming flat and hard nails. When these cells are away from and lose connections with the root, they die. This is why you do not get hurt when you trim your nails.

Nails grow continuously and these have to be trimmed regularly. Otherwise, they can grow extremely long, and become twisted like a run's horn.

Extend Your Understanding Know Zone)

Did you know that the thinnest skin is found on your eyelids? It is around 0.02 mm thick

Interactions with Other Body Systems

The human body is composed of many subsystems that work together to maintain their functions. The akia, for instance, works with the immune system, circulatory system, and nervous system to keep the body healthy.

The oil glands in the skin secrete an oily substance that increases the body's Iso contains immune system cells defense against infectious organisms that fight against germs

The integumentary system also works closely with the circulatory system by maintaining the temperature of the body through the capillaries-se fine blood vessels near the surface of the skin. The integumentary system also works with the nervous system to allow you to feel sensations

Care for the Integumentary System

In order to maintain the proper functioning of the integumentary system, you need to practice certain health habits.

Follow a healthy diet. Eat foods that are rich in protein such as milk, meat, and beans, as well as fruits and vegetables, which are good sources of vitamins and minerals

Practice safety precautions. Exercise safety in activities that may cause skin injuries such as burns and cuts. When needed, wear clothing such as gloves or socks to protect your skin.

Avoid too much exposure to the sun. proper diet

Depositing yourself too long to ultraviolet rays can cause skin damage or disease on your umbrella if you intend to stay long under the sun. Using sunblock is also needed. Your hair can also be damaged from exposure to too much sunlight

"lamage include color changes and broken or split erds.

avoiding sun exposure

Take a bath every day. Always keep your skin, hair, and nails clean to prevent illness-causing organisms from staying on them and eventually entering the body.

daily bath

enough steep

Always get enough rest and sleep. Stress or tension caused by pressure or problems can negatively affect the skin. Stress may cause the release of certain substances that can make the skin dry or damaged.

The integumentary system is a set of organs that provides the external bodily covering. Its main function is to protect everything in the body from any harm. It is composed of the skin, hair, and nails

The skin is an organ that is made up of a layer of tissues that protect underlying organs

The main layers of the skin are the epidermis, dermis, and hypodermis.

: Hair, the threadlike outgrowths from follicles found in the dermis, is made of a tough protein called keratin, which protects the hair from dryness and damage, and melanin, which gives hair its color Humans have three types of hair lanugo, vellus, and terminal hair

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Hair grows in cycles of three phases: growth, transitional, and resting

Phase

A nail is a keratinous covering for the tips of the fingers and toes

Nails consist of the nail plate, nail bed, cuticle, nail fold, lunula and the root

The integumentary system works with the immune system, circulatory system, and nervous system to keep the body healthy.

To care for the integumentary system, follow a healthy diet, practice safety precautions, take a bath every day, and get enough rest and

Sleep.

Express Your Learning

Answer the following questions

1. What is the integumentary system? What is its function?

2. What are the different parts of the integumentary system? What are the functions of each part?

3. What are the different ways of taking care of the integumentary system? List down three
4. How did you feel about the lesson? Explain your answer.

Lesson 3

Digestive System

Do List

1. Name the parts of the digestive system and their functions.
2. Identify the functions of the digestive system.
3. Explain how the organs of the digestive system work together with other organ systems
- 4 Enumerate ways of taking care of the digestive system.

Energize Your Mind

What food do you usually eat during recess? Do you know where the food you supply goes after you swallow it? Did you know that digestion of the food you eat begins in the mouth?

Equip Yourself

Think of one kind of food that you like. How do you eat the food? What parts of the body do you use when you eat? What happens to the food that you eat?

The digestion or processing of food begins immediately after it enters your mouth. This process is made possible by the digestive system.

The digestive system is a system of organs responsible for converting food into nutrients that are absorbed and used by the body to keep all its systems working. It is also responsible for the removal of wastes that cannot be converted, absorbed, and used by the body.

tarsals (ankle bones) are examples of short bones. The irregular bones have complex shapes that allow them to protect delicate internal organs. The vertebrae, which are the bones in the spinal column, protect the spinal cord. The spinal cord is a thick bundle of nerves that connects all the nerves of the body to the brain. The sesamoid bones are small, round bones that are found in the tendons of the body. Tendons are strong tissues that connect muscles to the bones to allow movement. Sesamoid bones protect the tendons from wear and tear. The patella, also known as the kneecap, is an example of a sesamoid bone. Can you identify the five types of bones in the images below?

Extend Your Understanding (Word Unlock)

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The cell is the basic structural and functional unit of all living things.

A tissue is a group of similar cells that work in a coordinated manner to perform a specific function.

An organ is a group of tissues that serve a common function.

Joints

The articular surface, commonly known as the joint, is the connection between bones in the body. The connections form a wholly functioning skeletal system. The joints allow different kinds and degrees of bodily movements with the help of muscles and bones. Without the joints, movement will not be possible.

Joints are classified according to the degree of their movement.

Equip Yourself

As you move, many parts of your body are involved. These parts work together and make up the musculoskeletal system. The musculoskeletal system is an organ system that allows the body to move using the muscular and skeletal systems. It provides the form, support, and stability of the body. It is composed of bones, joints, and muscles. It is also composed of tendons, ligaments, and cartilages.

Parts and Functions of the Musculoskeletal System

Bones

A bone is a living organ. It is made mostly of collagen, a protein that provides its soft framework. The mineral calcium phosphate gives strength and hardness to this framework. The bones make up the skeletal system of the body. At birth, the human body has over 300 bones. Many of these bones fuse together during development. The known number of separate bones is 206, but this does not count the numerous small bones. Thus, nobody really knows the exact number of bones in the human body.

Light in weight yet strong and hard, bones

cancellous bones-

cortical bones

have a complex structure and come in various shapes and sizes. The largest bone of the human body is the thigh bone or femur. The smallest bone is the stirrup in the middle ear. Bones are classified according to type of tissue: the cortical bone and the cancellous bone.

The cortical bone, also called compact bone, is the hard exterior that is strong and dense. It provides structural support to the body and protects internal organs. It also supports movement and releases calcium and other elements. Most of the body's calcium is stored in the bones and teeth.

The cancellous bone, also called the spongy bone, makes up the inner layer of the bone. It contains the bone marrow where blood cells are produced.

There are five types of bones: flat, long, short, irregular, and sesamoid.

such as the brain and the heart. The long bones provide strength, structure, and mobility. Some examples of long bones include the thigh bone, the humerus, the long organs shaped short bones provide stability and movement. The carpals (wrist bones) and one in the arm, and the radius and the ulna, the bones in the forearm. The cube

Equip Yourself

All organisms have the ability to detect changes within themselves and in their environments. Their survival depends on how well they could respond to changes in their surroundings. One factor that is necessary for survival is a speedy reaction or response. Reactions or responses involve the nervous system. The nervous system is a complex network of nerves and specialized cells that transmit messages between different parts of the body. A nerve is a cordlike bundle of fibers that connects the nervous system with other parts of the body.

Main Components of the Nervous System

The nervous system has two main components: the central nervous system and the peripheral nervous system.

Central Nervous System

The central nervous system (CNS) is made up of the brain and the spinal cord. It is called "central" because it receives information from the environment and the body and coordinates and controls thinking, emotions, memory, breathing, heart rate, body temperature, and other processes and activities of the body.

to peripheral nervous system

to peripheral nervous system

central nervous system

Neurons

dendrite

axon

cell body

Neurons are the cells that make up the brain and the nervous system. A neuron, or nerve cell, is the basic functional unit of the brain and is designed to transmit information to other nerve cells. A neuron consists of a cell body, dendrites, and an axon. A cell body contains the nucleus and connects to the dendrites and axons. Dendrites are tube-like branches that receive information from other neurons and transmit them to the cell body. An axon, which extends from the cell body and generates smaller branches that end at nerve terminals, carries information away from the cell body to another neuron.

neurons

There are three types of neurons. These are the sensory neurons, motor neurons, and interneurons. Sensory neurons are nerve cells that are activated by input from the environment through the senses. When you touch a hot object or smell something burning, these neurons send signals to the nervous system, which processes the information. Motor neurons carry messages from the brain, to the spinal cord, and then to the skeleton and muscles. Interneurons create neural pathways that connect sensory and motor neurons. An example of a neural pathway is seen in the knee-jerk reflex. When a spot on the knee is tapped with a rubber hammer, receptors send a signal to the spinal cord through the sensory neuron, which passes the information to the motor neuron. Impulses travel to the motor neuron, stimulating the leg muscle to contract and resulting in a quick response, the knee jerk.

Get a glimpse of how neurons look by doing the activity below.

Explore Around

MODEL ME A NEURON

What You Need

picture of a neuron

recyclables (plastic cups or bottles)

sheets of colored paper illustration board

glue

scissors

What to Do

1. Using the picture of a neuron as guide, draw the different parts of a neuron in the plastic cup.
2. Cut the different parts.
3. Cover the parts with the colored paper. Let each part have a specific color.
4. Assemble the parts. Then glue them on the illustration board.

What You Have Learned

Answer the following questions. Write your answers in the space provided.

1. What are the parts of a neuron?
2. How would you compare dendrites with axons?

How You Learned

1. How did the activity make sense to you?
2. Which part of the lesson helped you in performing the activity?

Brain

The brain is a highly specialized organ in the nervous system. It is the most complex and one of the largest organs in the human body. The human brain is a soft, grayish white structure that is protected by the skull and a layer of tissue. On the brain's surface are bright red arteries and bluish veins. The brain is made up of billions of nerves that communicate in trillions of connections called synapses. It is divided into three main parts: the cerebrum, the cerebellum, and the brain stem.

cerebrum

brain stem

cerebellum

the brain

The cerebrum is the largest part of the brain. It is responsible for your thoughts and actions and is thus considered the thinking part of the brain.

responsible for coordination of body movements and maintaining body posture and balance.

The brain stem is found in the lowest part of the brain. It controls basic body functions such as swallowing, breathing, and heart rate. Ban

Extend Your Understanding (Know Zone)

The cerebrum consists of two hemispheres. The left hemisphere controls the right side of the body and is responsible for right-handedness and tasks that involve logic and analytical thought, such as in mathematics and science. The right hemisphere controls the left side of the body and is responsible for left-handedness and performs tasks that involve creativity, imagination, and other abstract thoughts such as colors, shapes, and music.

analysis

logic

facts

sequencing

mathematics

language
creativity
Intuition
feelings
Imagination
daydreaming
arts
Spinal Cord
spinal cord.

The spinal cord is a bundle of nerve tissue that links the brain with the rest of the body. Protected by the vertebral column that runs about the full length of the back, the spinal cord serves as a pathway for the exchange of information and signals between the body and the brain. The center of the spinal cord contains gray matter, which is surrounded by white matter. Gray matter is composed mainly of numerous cell bodies of neurons. Meanwhile, white matter is generally made of axons. Spinal nerves originate in the spinal cord and form part of the peripheral nervous system.

the spinal cord.

Peripheral Nervous System

The peripheral nervous system (PNS) consists of the nerves and nerve clusters outside the brain and the spinal cord, extending to the other parts of the body. There are 12 pairs of cranial nerves that are attached to the brain and 31 pairs of spinal nerves attached to the spinal cord. The primary function of the PNS is to connect the central nervous system and the body tissues. It allows the brain and the spinal cord to receive and send information to the body and allows you to react or respond to stimuli in the environment.

Care for the Nervous System

Since the nervous system manages the whole body, it is important for it to be free from stress and illnesses. Here are some ways of caring for this organ system.

Be aware of what you eat. To take care of the nervous system, you have to be aware of what you eat and drink. As with the other body systems, what you eat directly affects the different organs and nerves that make up the nervous system.

Have a balanced diet. You should always have a balanced diet. A good diet should include fresh fruits and vegetables. It should also contain foods that are rich in vitamins B6, B12, and B9 (folic acid), which protect the health of the nervous system.

Take plenty of rest. You should take plenty of rest to boost your body's defense system against diseases. During sleep, tissues are repaired, and hormones that help the nervous system and other body systems develop are secreted during this time.

Rest your eyes. Do not abuse it by watching too much TV, using handheld devices for a long time, or reading in poorly lighted places. When closing your eyes to rest, it is best to turn off or block all sources of light from your eyes. This will keep your optic nerves from being disturbed by reflected light from objects around you.

Take care of your ears. Just like your eyes, the ears are crucial for experiencing the world around you and relaying information to the brain for processing. Avoiding loud sounds or turning up the volume of your music player can damage your ears.

substances. Cigarettes contain harmful substances that may damage the nervous system.

Mind-altering drugs cause irreversible damage to your brain and the rest of your nervous system and should thus be avoided.

Exercise regularly. Exercising helps keep your body, especially your nervous system, healthy and fit.

Protect yourself from head and back injuries. You can keep your brain and spinal cord protected by wearing protective gear, such as a helmet and back protector when doing activities such as biking and by fastening your seatbelt when riding a vehicle.

Interaction with Other Systems

The nervous system receives and relays signals throughout the body. Just like the central processing unit of a computer, the brain tries to analyze what goes on in one part of the body based on the signals that it receives. The brain analyzes and makes the necessary adjustments. The reflex action of the musculoskeletal system to pain is an example of such an interaction. Another example is when the weather gets too cold. Your brain tells your body to begin shivering, which is the body's way of getting warm. Or when your body becomes too hot because of the weather or after a vigorous exercise, your brain tells your body to sweat in order to cool down. Ravidas

The nervous system is a complex network of nerves and specialized cells that transmit messages between different parts of the body.

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A nerve is a cordlike bundle of fibers that connects the nervous system with other parts of the body.

The nervous system has two main components:

- > The central nervous system (CNS) is made up of the brain and spinal cord.

- The peripheral nervous system (PNS) consists of the nerves and nerve clusters outside the brain and the spinal cord, extending to the other parts of the body.

A neuron or nerve cell is the basic functional unit of the brain that is designed to transmit information to other nerve cells.

A neuron consists of

A cell body, which contains the nucleus and connects to the dendrites and axons;

- > Dendrites, which are tube-like branches that receive information from other neurons and transmit them to the cell body; and

An axon, which extends from the cell body and generates smaller branches that end at nerve terminals, carries information away from the cell body to another neuron.

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Sensory neurons are nerve cells that are activated by input from the environment through the senses.

➤ Motor neurons carry messages from the brain, to the spinal cord, and then to the skeleton and muscles.

Interneurons create neural pathways that connect sensory and motor neurons.

The brain is a soft, grayish white structure that is protected by the skull and a layer of tissue.

The brain is divided into three main parts.

➤ The cerebrum is the largest part of the brain and is responsible for your thoughts and actions.

The cerebellum is responsible for coordination of body movements and maintains body posture and balance.

The brain stem controls basic body functions such as swallowing, breathing, and heart rate.

The spinal cord is a bundle of nerve tissue that links the brain with the rest of the body.

Express Your Learning

Answer the following questions:

1. How does the central nervous system compare with the peripheral nervous system in terms of their functions?

2. What are the types of neurons? Describe each

3. What are some ways of taking care of the nervous system?

How do the organs of the nervous system work together?

5. How do you feel about the lesson? Explain your answer.

Lesson 7

Excretory System

Do List

1. Name the parts of the excretory system.
2. Identify the functions of each part of the excretory system.
3. Enumerate ways of taking care of the excretory system.
4. Explain how the major organs of the excretory system work together with other organ systems.

Energize Your Mind

You have often been told to drink at least eight glasses of water a day. During the hot summer months, you drink all the time! Young people love to drink prepared juice, soda, or milk tea more than they love to drink water. Why is that so? Have you ever thought about how what you drink

is used by your body? Do you know how water and unwanted chemicals in the body are eliminated?

Equip Yourself

The body has its own filtration system. It has the ability as well as the mechanisms to expel all waste materials that it generates and the toxins that are present in it. In humans, water, prepared juice, soda, milk tea, and any kind of drink are converted in the body and absorbed, and waste materials are eliminated from the body.

The excretory system is a body system that is responsible for the removal of wastes, toxins, and excess and unnecessary materials from the body. This is essential in maintaining internal chemical processes and preventing damage to the body. Carbon dioxide is released through exhalation, and food waste is evacuated as feces.

Parts of the Excretory System and Their Functions

The organs that form the excretory system include the kidneys, ureter, urinary bladder, and the urethra.

Kidneys

The kidneys are two bean-shaped organs located on either side of the spine, below the ribs, and behind the belly. While each kidney is about the size of a fist, it does the monumental job of filtering the blood with the help of tiny filters called nephrons.

The kidneys clean the bloodstream by eliminating waste materials that would otherwise cause damage to the body if kept in the blood. The liquid by-product of this filtration process is urine. Urine is excreted, whereas the filtered blood goes back into the body. The kidneys are also important in maintaining the volume of fluid in the body and in maintaining the chemical conditions of a healthy body.

Ureters

There are two ureters in the body, one attached to each kidney. A ureter is a long, narrow tubular passageway about 30 cm long. Each kidney has its own ureter through which urine passes. Urine does not flow in a steady stream. The muscles along the walls of the ureters contract, squirting urine toward the bladder where it is stored before elimination. The way the ureters are connected to the bladder wall prevents urine from flowing back to the body.

Urinary Bladder

The urinary bladder is a hollow, muscular sac where urine from the ureters is stored. It is lined by layers of muscle tissue that stretch to hold urine. Urination is the process of expelling urine from the bladder. It is also called voiding of the bladder. The bladder is able to hold 400 to 600 ml of urine. During urination, the bladder muscles squeeze, and two valves allow the urine to flow out.

Urethra

Once the bladder's storage capacity is reached, the urinary bladder starts to contract, and urine exits from the bladder and is expelled to the outside through a tubular passageway called the urethra. The opening of the urethra is controlled by muscles that keep it shut. The urethra is longer in males than in females.

Care for the Excretory System

Since the excretory system deals with waste materials, you need to adopt simple habits to keep your body strong and healthy and avoid problems in your system.

Drink enough water

Drinking plenty of water will cleanse not only your kidneys and the rest of your excretory system but also your other body systems. Drink water whenever you feel thirsty to keep your body parts hydrated and working.

Be mindful about what you eat and drink

Eating healthy food helps keep your excretory system functioning properly. Frequent consumption of foods with high salt content, such as chips and instant noodles, will eventually harm your body. Drinks such as soda and milk tea are often loaded with sugar and artificial flavorings that your body will be forced to digest and absorb.

Watch your medicine intake

Medicines such as anti-allergy tablets, antibiotics, and pain relievers may do you more harm than good if you take them without a parent's or doctor's advice. Moreover, do not think that medicine that has helped your friend or sibling get well will do the same for you. Always seek the help of your parents and elders as well as the advice of a doctor.

Exercise regularly

Be active to keep your excretory organs healthy.

Avoid tobacco products, alcohol, and illegal substances

These often contain toxic chemicals that can seriously harm your excretory system.

Interaction with Other Systems

When the excretory system is not working well, the whole body system suffers. This is especially true if the kidneys are not functioning. The toxic materials that are accumulated in the blood can get transported to other systems and cause damage to the other organs.

For some scientists, the excretory system not only includes the kidney and the urinary organs but also consists of other body systems that expel waste materials in different forms. It includes the respiratory system, which exhales gases from the body, the integumentary system, which flushes out excess water and some salts through the skin, and the digestive system, which removes waste materials in solid form. That is why all your body parts are interrelated. When an organ is in good condition, the other organs also stay healthy.

The excretory system is a body system that is responsible for the removal of wastes, toxins, and excess and unnecessary materials from the body to help maintain internal chemical processes and prevent damage to the body.

The excretory system includes the kidneys, ureters, urinary bladder, and urethra.

The kidneys are two bean-shaped organs that filter the blood.

The ureters are long, narrow tubular passageways through which urine passes from the kidney to the urinary bladder.

The urinary bladder is a hollow muscular sac where urine from the ureters is stored.

Urination is the process of excreting urine from the bladder.

Urine exits from the bladder and is expelled through a tubular passageway called the urethra.

Simple habits to keep the excretory system strong and healthy include drinking enough water, being mindful about what you eat and drink, watching your medicine intake, exercising regularly, and avoiding tobacco products, alcohol, and illegal substances.

Nervous System

Equip Yourself

All organisms have the ability to detect changes within themselves and in their environments. Their survival depends on how well they could respond to changes in their surroundings. One factor that is necessary for survival is a speedy reaction or response. Reactions or responses involve the nervous system. The nervous system is a complex network of nerves and specialized cells that transmit messages between different parts of the body. A nerve is a cordlike bundle of fibers that connects the nervous system with other parts of the body.

Main Components of the Nervous System

The nervous system has two main components: the central nervous system and the peripheral nervous system.

Central Nervous System

The central nervous system (CNS) is made up of the brain and the spinal cord. It is called "central" because it receives information from the environment and the body and coordinates and controls thinking, emotions, memory, breathing, heart rate, body temperature, and other processes and activities of the body.

Neurons

Neurons are the cells that make up the brain and the nervous system. A neuron, or nerve cell, is the basic functional unit of the brain and is designed to transmit information to other nerve cells.

A neuron consists of a cell body, dendrites, and an axon. A cell body contains the nucleus and connects to the dendrites and axons. Dendrites are tube-like branches that receive information from other neurons and transmit them to the cell body. An axon, which extends from the cell body and generates smaller branches that end at nerve terminals, carries information away from the cell body to another neuron.

There are three types of neurons. These are the sensory neurons, motor neurons, and interneurons.

Sensory neurons are nerve cells that are activated by input from the environment through the senses. When you touch a hot object or smell something burning, these neurons send signals to the nervous system, which processes the information.

Motor neurons carry messages from the brain, to the spinal cord, and then to the skeleton and muscles.

Interneurons create neural pathways that connect sensory and motor neurons. An example of a neural pathway is seen in the knee-jerk reflex.

Brain

The brain is a highly specialized organ in the nervous system. It is the most complex and one of the largest organs in the human body. The human brain is a soft, grayish white structure that is protected by the skull and a layer of tissue. The brain is made up of billions of nerves that communicate in trillions of connections called synapses. It is divided into three main parts: the cerebrum, the cerebellum, and the brain stem.

The cerebrum is the largest part of the brain. It is responsible for your thoughts and actions and is thus considered the thinking part of the brain.

The cerebellum is responsible for coordination of body movements and maintaining body posture and balance.

The brain stem is found in the lowest part of the brain. It controls basic body functions such as swallowing, breathing, and heart rate.

Spinal Cord

The spinal cord is a bundle of nerve tissue that links the brain with the rest of the body. Protected by the vertebral column that runs about the full length of the back, the spinal cord serves as a pathway for the exchange of information and signals between the body and the brain. The center of the spinal cord contains gray matter, which is surrounded by white matter. Gray matter is composed mainly of numerous cell bodies of neurons. Meanwhile, white matter is generally made of axons. Spinal nerves originate in the spinal cord and form part of the peripheral nervous system.

Peripheral Nervous System

The peripheral nervous system (PNS) consists of the nerves and nerve clusters outside the brain and the spinal cord, extending to the other parts of the body. There are 12 pairs of cranial nerves that are attached to the brain and 31 pairs of spinal nerves attached to the spinal cord. The primary function of the PNS is to connect the central nervous system and the body tissues. It allows the brain and the spinal cord to receive and send information to the body and allows you to react or respond to stimuli in the environment.

Care for the Nervous System

Since the nervous system manages the whole body, it is important for it to be free from stress and illnesses. Here are some ways of caring for this organ system.

Be aware of what you eat

To take care of the nervous system, you have to be aware of what you eat and drink. As with the other body systems, what you eat directly affects the different organs and nerves that make up the nervous system.

Have a balanced diet

You should always have a balanced diet. A good diet should include fresh fruits and vegetables. It should also contain foods that are rich in vitamins B6, B12, and B9 (folic acid), which protect the health of the nervous system.

Take plenty of rest

You should take plenty of rest to boost your body's defense system against diseases. During sleep, tissues are repaired, and hormones that help the nervous system and other body systems develop are secreted during this time.

Rest your eyes

Do not abuse it by watching too much TV, using handheld devices for a long time, or reading in poorly lighted places.

Take care of your ears

Just like your eyes, the ears are crucial for experiencing the world around you and relaying information to the brain for processing. Avoid loud sounds or turning up the volume of your music player can damage your ears.

Exercise regularly

Exercising helps keep your body, especially your nervous system, healthy and fit.

Protect yourself from head and back injuries

You can keep your brain and spinal cord protected by wearing protective gear, such as a helmet and back protector when doing activities such as biking and by fastening your seatbelt when riding a vehicle.

Interaction with Other Systems

The nervous system receives and relays signals throughout the body. Just like the central processing unit of a computer, the brain tries to analyze what goes on in one part of the body

based on the signals that it receives. The brain analyzes and makes the necessary adjustments. The reflex action of the musculoskeletal system to pain is an example of such an interaction.

Another example is when the weather gets too cold. Your brain tells your body to begin shivering, which is the body's way of getting warm. Or when your body becomes too hot because of the weather or after a vigorous exercise, your brain tells your body to sweat in order to cool down.

The nervous system is a complex network of nerves and specialized cells that transmit messages between different parts of the body.

A nerve is a cordlike bundle of fibers that connects the nervous system with other parts of the body.

The nervous system has two main components:

- The central nervous system (CNS) is made up of the brain and spinal cord.
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A neuron or nerve cell is the basic functional unit of the brain that is designed to transmit information to other nerve cells.

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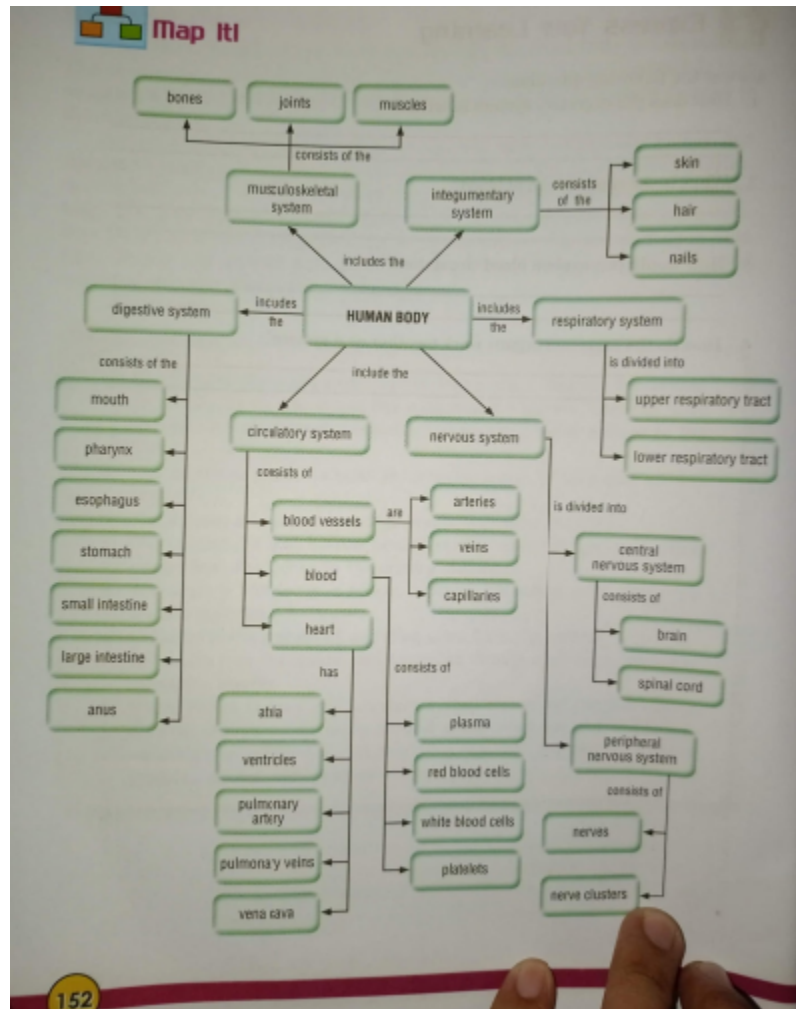
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Wrap Up

The musculoskeletal system consists of the bones, joints, and muscles.

A bone is a living organ that is mostly made up of collagen and calcium phosphate.

Joints are the connections between the bones in the body.

Muscles are the specialized tissues of the body that have the ability to contract and effect movement.

Some ways to care for the musculoskeletal system include exercising, getting the right food and nutrition, and getting enough sleep.

The integumentary system is a set of organs that provides the external bodily covering. Its main function is to protect everything in the body from any harm. It is composed of the skin, hair, and nails.

The skin is an organ that is made up of a layer of tissues that protects underlying organs.

Hair are the threadlike outgrowths from the follicles found in the dermis.

Nails are continuous coverings for the tips of the fingers and toes.

To care for the integumentary system, follow a healthy diet, practice safety precautions, take a bath every day, and get enough rest and sleep.

The digestive system is a system of organs that is responsible for converting food into nutrients that are absorbed and used by the body to keep all its systems working. It is also responsible for the removal of food that cannot be converted, absorbed, or used by the body.

Food can be digested through a combination of two methods: mechanical digestion and chemical digestion.

The digestive tract or alimentary canal is the continuous muscular tube that carries food through different parts of the digestive system.

The circulatory system is a group of organs or structures that is responsible for circulating blood and transporting nutrients and oxygen throughout the body. It also removes substances and waste products that are not needed by the body.

The human circulatory system is composed of three major parts: blood vessels, blood, and heart.

Blood vessels are a series of hollow tubes through which blood passes.

The blood is a body fluid that brings nutrients, oxygen, and other substances to the cells and transports waste materials away from those cells.

The heart is a muscular organ that pumps blood throughout the body.

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